Chapter 17

Fuzzy Clustering

The **fuzzy clustering** is considered as soft clustering, in which each element has a probability of belonging to each cluster. In other words, each element has a set of membership coefficients corresponding to the degree of being in a given cluster.

This is different from k-means and k-medoid clustering, where each object is affected exactly to one cluster. K-means and k-medoids clustering are known as hard or non-fuzzy clustering.

In fuzzy clustering, points close to the center of a cluster, may be in the cluster to a higher degree than points in the edge of a cluster. The degree, to which an element belongs to a given cluster, is a numerical value varying from 0 to 1.

The **fuzzy c-means** (FCM) algorithm is one of the most widely used fuzzy clustering algorithms. The centroid of a cluster is calculated as the mean of all points, weighted by their degree of belonging to the cluster:

In this article, we'll describe how to compute fuzzy clustering using the R software.

17.1 Required R packages

We'll use the following R packages: 1) *cluster* for computing fuzzy clustering and 2) *factoextra* for visualizing clusters.

17.2 Computing fuzzy clustering

The function *fanny*() [*cluster* R package] can be used to compute fuzzy clustering. **FANNY** stands for **fuzzy analysis clustering**. A simplified format is:

```
fanny(x, k, metric = "euclidean", stand = FALSE)
```

- **x**: A data matrix or data frame or dissimilarity matrix
- **k**: The desired number of clusters to be generated
- **metric**: Metric for calculating dissimilarities between observations
- **stand**: If TRUE, variables are standardized before calculating the dissimilarities

The function *fanny*() returns an object including the following components:

- **membership**: matrix containing the degree to which each observation belongs to a given cluster. Column names are the clusters and rows are observations
- **coeff**: Dunn's partition coefficient F(k) of the clustering, where k is the number of clusters. F(k) is the sum of all squared membership coefficients, divided by the number of observations. Its value is between 1/k and 1. The normalized form of the coefficient is also given. It is defined as $(F(k) - 1/k)/(1 - 1/k)$, and ranges between 0 and 1. A low value of Dunn's coefficient indicates a very fuzzy clustering, whereas a value close to 1 indicates a near-crisp clustering.
- **clustering**: the clustering vector containing the nearest crisp grouping of observations

For example, the R code below applies fuzzy clustering on the USArrests data set:

```
library(cluster)
df <- scale(USArrests)     # Standardize the data
res.fanny <- fanny(df, 2)  # Compute fuzzy clustering with k = 2
```

The different components can be extracted using the code below:

```
head(res.fanny$membership, 3) # Membership coefficients
```

```
##              [,1]      [,2]
## Alabama 0.6641977 0.3358023
## Alaska  0.6098062 0.3901938
## Arizona 0.6862278 0.3137722
```

```
res.fanny$coeff # Dunn's partition coefficient
```

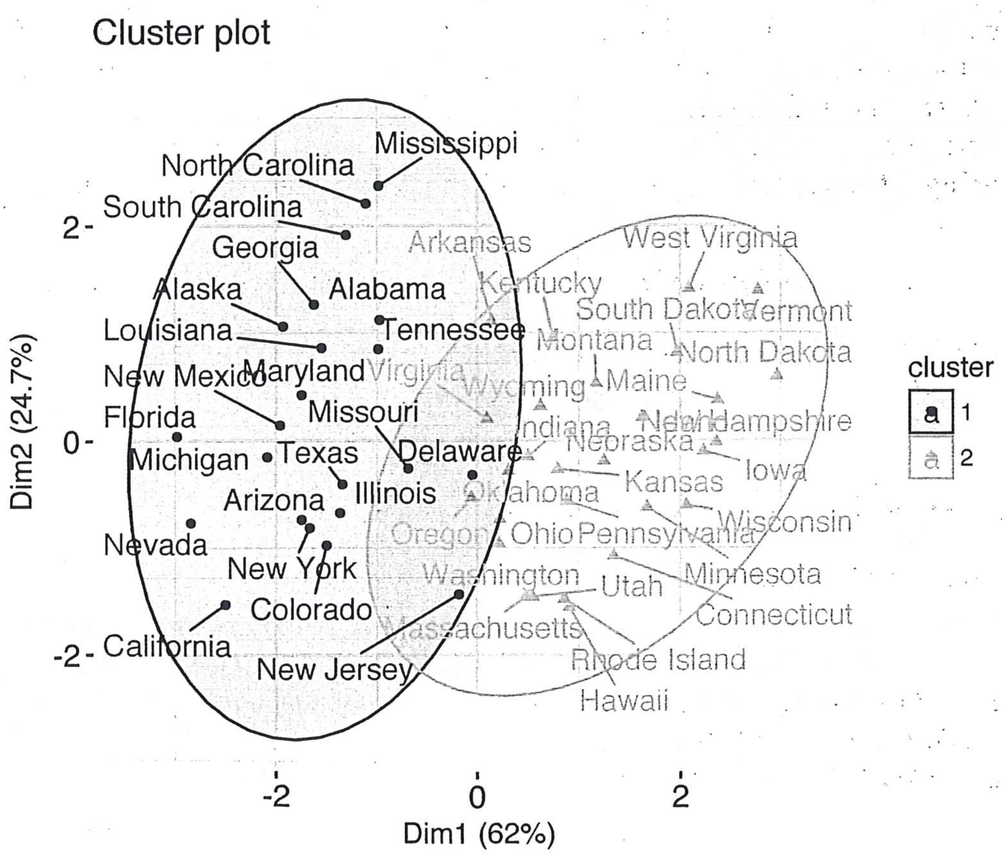
```
## dunn_coeff normalized
##  0.5547365  0.1094731
```

```
head(res.fanny$clustering) # Observation groups
```

```
##    Alabama      Alaska     Arizona    Arkansas California   Colorado
##          1           1           1           2          1          1
```

To visualize observation groups, use the function *fviz_cluster*() [*factoextra* package]:

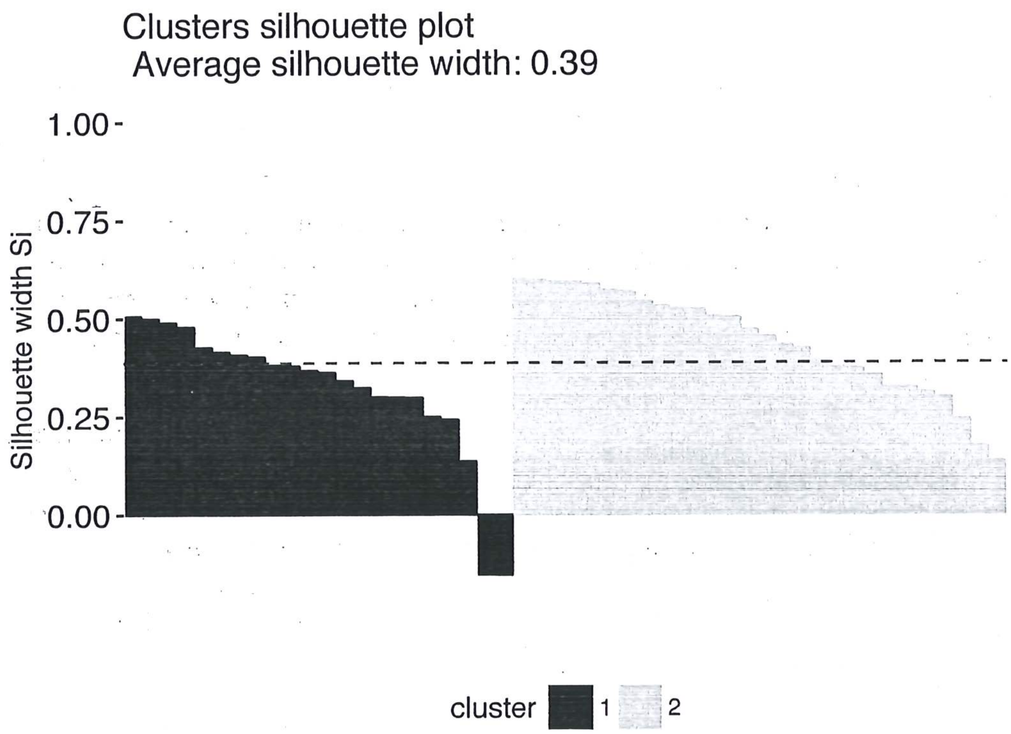
```
library(factoextra)
fviz_cluster(res.fanny, ellipse.type = "norm", repel = TRUE,
             palette = "jco", ggtheme = theme_minimal(),
             legend = "right")
```



To evaluate the goodnesss of the clustering results, plot the silhouette coefficient as follow:

```
fviz_silhouette(res.fanny, palette = "jco",
                ggtheme = theme_minimal())
```

```
##   cluster size ave.sil.width
## 1       1   22          0.32
## 2       2   28          0.44
```



17.3 Summary

Fuzzy clustering is an alternative to k-means clustering, where each data point has membership coefficient to each cluster. Here, we demonstrated how to compute and visualize fuzzy clustering using the combination of *cluster* and *factoextra* R packages.